

Kevin Cimorelli
Site Vice President

Susquehanna Nuclear, LLC
769 Salem Boulevard
Berwick, PA 18603
Tel. 570.542.3795 Fax 570.542.1504
Kevin.Cimorelli@TalenEnergy.com



12/11/2019

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387(388)/2019-003-00
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7828

Docket No. 50-387
and 50-388

Attached is Licensee Event Report (LER) 50-387(388)/2019-003-00. This LER reports a condition in which the D Diesel Generator was slow to start and reach required frequency. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications and 10 CFR 50.73(a)(2)(v) as an event or condition which could have prevented the fulfillment of a safety function. There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-387(388)/2019-003-00

Copy: NRC Region I
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Ms. S. Goetz, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (04-2018)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020						
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)									
1. Facility Name Susquehanna Steam Electric Station Unit 1					2. Docket Number 05000387		3. Page 1 OF 3				
4. Title D Diesel Generator Inoperable due to Inadequate Post-Maintenance Test											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number	
10	13	2019	2019	003	00	12	11	2019	Susquehanna Steam Electric Station Unit 2	05000388	
									Facility Name	Docket Number	
										05000	
9. Operating Mode		11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)									
1		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
100		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☒ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☒ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)	
		☐ 20.2203(a)(2)(vi)			☒ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(iii)	
		☐ 50.73(a)(2)(i)(C)						☐ Other (Specify in Abstract below or in NRC Form 366A)			
12. Licensee Contact for this LER											
Licensee Contact Shane Jurek, Senior Engineer – Nuclear Regulatory Affairs								Telephone Number (Include Area Code) (570) 542-1695			
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES		
D	EK	DG	C634	Y							
14. Supplemental Report Expected ☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No					15. Expected Submission Date			Month	Day	Year	
Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)											
On October 13, 2019, at 10:08, during performance of surveillance testing, the D Diesel Generator (DG) failed to reach the required frequency within the time specified by the acceptance criteria. Operators declared Surveillance Requirement 3.8.1.7 failed and entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, Condition B, on both Susquehanna Steam Electric Station units for the inoperable D DG. The other three DGs (A, B, and C) were verified to be OPERABLE before operators commenced a subsequent test of the D DG at 12:12, during which the D DG met all required acceptance criteria. The D DG was declared OPERABLE at 17:56 and LCO 3.8.1, Condition B, was exited for both units. The failure to reach the required frequency within the acceptance criteria was the result of inadequate oil pressure in the D DG mechanical governor internals which was caused by an inadequate Post Maintenance Test (PMT). The condition of the inoperable D DG was determined to have existed since September 17, 2019, and the event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS. Due to a concurrent inoperability of the C DG, the event was also determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v) as an event or condition which could have prevented the fulfillment of a safety function. Corrective actions include procedure updates to ensure adequate PMTs are performed in the future. There were no actual safety consequences associated with this event.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 1		05000-387	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	003	00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On October 13, 2019, at 10:08, operators started the D Diesel Generator (DG) [EIS Code/Component Code: EK/DG] in accordance with the applicable surveillance procedure. The D DG reached the required frequency at a time of 10.75 seconds, which exceeds the acceptance criteria of 10 seconds. Operators declared Surveillance Requirement (SR) 3.8.1.7 failed and entered Technical Specification (TS) Limiting Condition of Operation (LCO) 3.8.1, Condition B, on both Susquehanna Steam Electric Station (SSES) units for the inoperable D DG. The other three DGs (A, B, and C) were verified to be OPERABLE before operators commenced a subsequent test of the D DG at 12:12, during which the D DG met all required acceptance criteria. The D DG was declared operable at 17:56 and LCO 3.8.1, Condition B, was exited for both units.

Upon engine start, the engine cranked until the mechanical governor [EIS Code/Component Code: EK/65] lever arm was able to travel to the maximum fuel position. The mechanical governor is not designed to hold pressure during standby; since the engine is not operating, there is no pressure being generated in the mechanical governor. During the next engine start, the governor will need to build up internal pressure prior to being able to move the output lever arm back to the maximum fuel position.

On September 17, 2019, work was performed on the D DG which required draining the starting air receivers. Upon completion of the work, a Post-Maintenance Test (PMT) was performed to verify leak-tightness and proper operation of the Emergency Service Water System supply valve [EIS Code/Component Code: BI/V] to the intercoolers. However, no PMT DG run was performed. Absent a PMT DG run, the DG mechanical governor was not primed with governor oil. This condition rendered the D DG inoperable due to its impact on the ability to reach the required frequency within 10 seconds.

SSES TS LCO 3.8.1 requires, in part, that four DGs are OPERABLE in MODES 1, 2, and 3. Condition B requires that an inoperable DG be restored to OPERABLE status within 72 hours or that each unit be shut down to MODE 3 in 12 hours and MODE 4 in 36 hours. The D DG was inoperable from September 17, 2019, until it was declared OPERABLE at 17:56 on October 13, 2019. Both SSES units remained in MODE 1 during the 26-day period which exceeds the 72 hours allowed by Condition B. As a result, this condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

The C DG was also inoperable from 07:54 on October 2, 2019, through 23:40 on October 3, due to a fuel oil leak and again from 20:14 on October 7 through 01:16 on October 8 during DG swap following repairs to the

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CONTINUATION SHEET**

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Susquehanna Steam Electric Station Unit 1		05000-387	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	003	00

leak. SSES TS LCO 3.8.1, Condition E requires that, with two or more DGs inoperable, at least three DGs be restored to an OPERABLE status within two hours or that each unit be shut down to MODE 3 in 12 hours and MODE 4 in 36 hours. The C and D DGs were concurrently inoperable for a period of approximately 40 hours which exceeds the 2 hours allowed by Condition E (the second concurrent inoperability lasted approximately five hours, which is less than the 2 hours plus 12 hours allotted by Conditions E and F). Furthermore, with the C and D DG concurrently inoperable, the condition was also determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v) as an event or condition which could have prevented the fulfillment of a safety function.

CAUSE OF EVENT

The cause of the slow start was an inadequate PMT after a starting air clearance resulted in depressurization of the governor oil. Work on the D DG on September 17, 2019, required control air to be taken out of service, thereby allowing pressure to bleed out of the mechanical governor until the next D DG start on October 13, 2019. Without a PMT DG run, the oil pressure was not restored in the DG mechanical governor.

ANALYSIS/SAFETY SIGNIFICANCE

The safety function of the DGs is to provide power to equipment as assumed in the safety analyses described in Chapter 15 of the Susquehanna Updated Final Safety Analysis Report. In this case, the accident analyses assume the DGs start within 25.1 seconds of a valid start signal. The start time of 10.75 seconds observed during the October 13, 2019, test is within that assumed in the accident analyses. Therefore, the safety function of the D DG was maintained and, despite the concurrent inoperability of the C and D DGs, three DGs were always available to perform their safety function.

Based on engineering analysis, this event will not be counted as a safety system functional failure under the NRC Reactor Oversight Process Performance Indicator.

CORRECTIVE ACTIONS

Corrective actions include procedure changes to require a PMT DG run whenever the starting air system is removed from service.

PREVIOUS OCCURRENCES

LER 50-387(388)/2016-008-00, "Inoperability of Diesel Generator Due to Misalignment of MOC [Mechanism-Operated Cell] Switch Contacts Due to Inadequate Post Maintenance Testing," dated May 19, 2016

LER 50-387(388)/2015-001-00, "Inoperability of the 'B' Emergency Diesel Generator due to Fuel Oil Leakage," dated April 25, 2015

LER 50-387(388)/2011-004-00, "'C' Emergency Diesel Generator Inoperable," dated February 13, 2012